NYU TANDON SCHOOL OF ENGINEERING

Anatomy of a Signal: Dissecting Non-Linear Factor Efficacy in US Equities

Arnav Sinha (as19390)

FRE 7801 Real-time Risk Management – Professor Naresh Malhotra

Setup of the Project

Problem Statement

In the era of the 'Factor Zoo,' where researchers start with thousands of noisy indicators, can modern non-linear Machine Learning distill true idiosyncratic alpha?

Data Universe (The Input) - High-Dimensional Panel Data

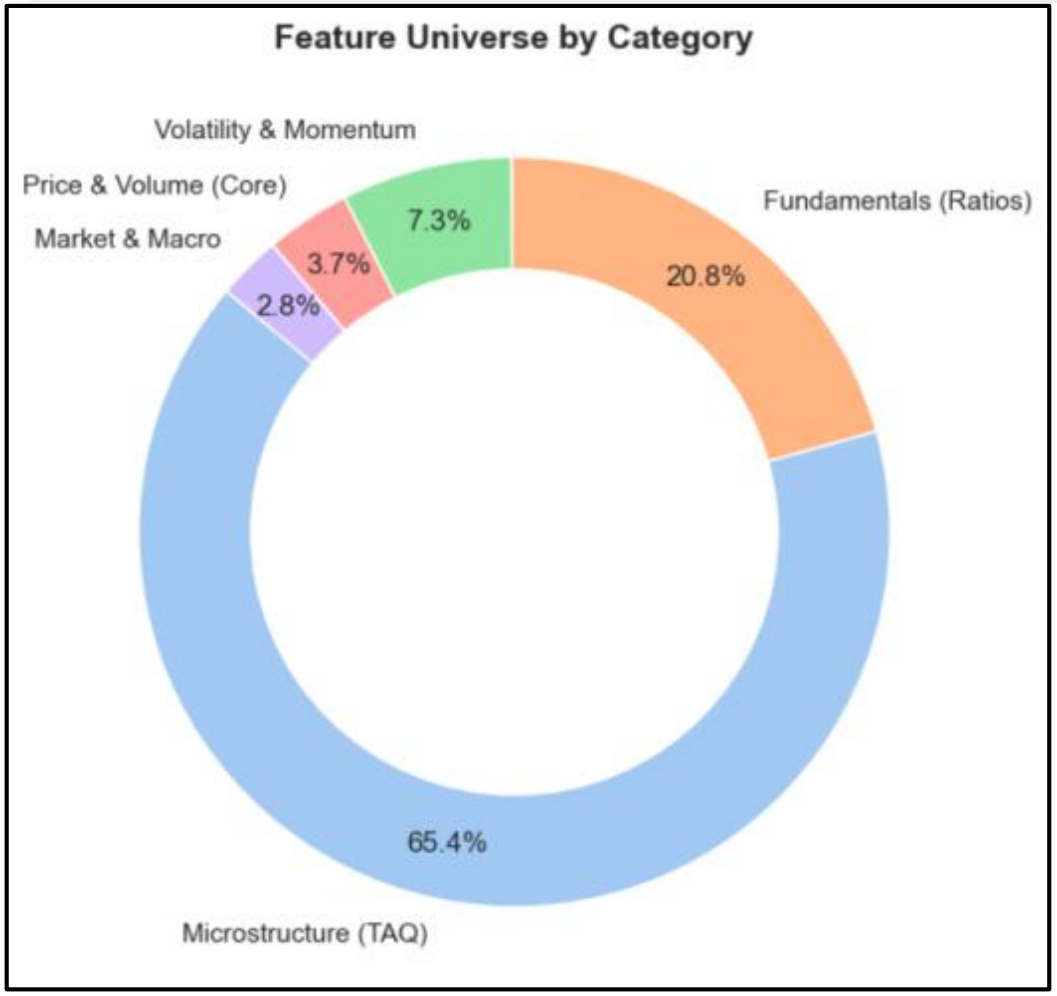
Bloomberg, WRDS. 494 Static S&P 500 Constituents (Daily) | Period: Jan 2005 – Dec 2014
Feature Matrix: 327 Features per stock, normalized daily (z-score). Microstructure: Tick-level granular data. Fundamentals: Accounting ratios. Technicals: Volatility, Momentum, RSI, Volume
Total Volume: ~1.1 Million Daily Observations.

Predictive Model & Portfolio Construction Framework

Model: LightGBM (Gradient Boosting Machine) | **Objective:** Predict forward daily returns r_{t+1}
Training: Rolling window approach to prevent look-ahead bias

Portfolio Construction: Simple Ranking. Allocation: Long Top 10% / Short Bottom 10%
Weighting: Dollar Neutral $\sum w_{\text{long}} = \sum w_{\text{short}}$ Beta Neutral: $\sum_i w_i \ \beta_i = 0$ **Friction:** Transaction costs applied at **4 bps (0.04%)** per turnover.

A peek into the Data



Objective: Construct a Balanced Panel to ensure stable model training.

Selection Criteria: Top 100 stocks by Market Cap with complete, continuous trading data (Jan 2005 – Dec 2014).

Trade-off: We prioritized computational stability and time-series continuity over a dynamic universe, ***accepting known Survivorship Bias*** to isolate the signal efficacy of the model itself.

Category	Count	Examples
Microstructure (TAQ)	214	Bid-ask spreads, BidLow…
Fundamentals (Ratios)	68	pe_, debt, cash...
Volatility & Momentum	24	Vol_, Momentum_, IVol...
Price & Volume (Core)	12	Close, Open, High...
Market & Macro	9	SPX, VIX, USGG3M...

Preprocessing of Data

Bloomberg Wide Format. WRDS: Panel/Long Format - Temporal Alignment ("As-Of" Merging)

"Dead Column" Removal, Type Casting, Look-ahead bias proof imputation strategy, no global imputation, no overfilling of data – Light GBM Handles missing data

Methodology: A Deep Dive

The Model: LightGBM (Gradient Boosting)
Mechanism: An ensemble of decision trees trained sequentially. Unlike Random Forest, each new tree explicitly targets the residuals (errors) of the previous trees.
Why LightGBM?

- *Efficiency*: Optimized for large dense matrices
- *Interaction Detection:* It automatically learns non-linear "If-Then" logic ("If Volume is High AND Volatility is Low, then Buy"). Linear regression misses this.

Training Protocol: Expanding Window (Walk-Forward Validation) to prevent look-ahead bias.

Scoring & Ranking
Input: 327 Features for Stock i on Day t.
Output: Raw Prediction Score ($\hat{y}$). A continuous number representing expected excess return.
Ranking: Every day in the Test Set (2013–2014), we rank the 100 stocks by their Score.

- Rank 1 = Highest conviction Long.
- Rank 100 = Highest conviction Short.

Eval Metric: The Information Coefficient (IC)
Definition: The Spearman Rank Correlation between Today's Scores and Tomorrow's Returns.

$$IC_t = Corr(Rank(\hat{y}_t), Rank(y_{t+1}))$$

Target: We don't need 90% accuracy. An IC of 0.03 to 0.05 (predicting the rank slightly better than random) is sufficient for profitability due to the law of large numbers.

Metric	In-Sample	Out-of-Sample
RMSE (Error)	0.018	0.022
R-Squared	4.2%	0.3%
Directional Accuracy	53.8%	51.4%
IC	0.054 *(Model fits the past well).*	0.021 *(Performance drops but remains positive)*

Portfolio Construction Math

The Naive Approach (The "Trap") vs The Strict Approach (The "Reality")

A. Naive

Selection: Long Top 10 stocks / Short Bottom 10 stocks.

Weighting: Dollar Neutral ($\sum w_i = 0$). Equal weight (10%) per stock.

The Flaw: It ignores Risk Factors. If the model picks 10 Tech stocks (High Beta) and shorts 10 Utility stocks (Low Beta), the portfolio is Net Long Beta.

B. Strict

Optimization Problem:

$\text{maximize } \alpha^T w$ subject to:

$$\sum w_i = 0 (Dollar\ Neutral)$$

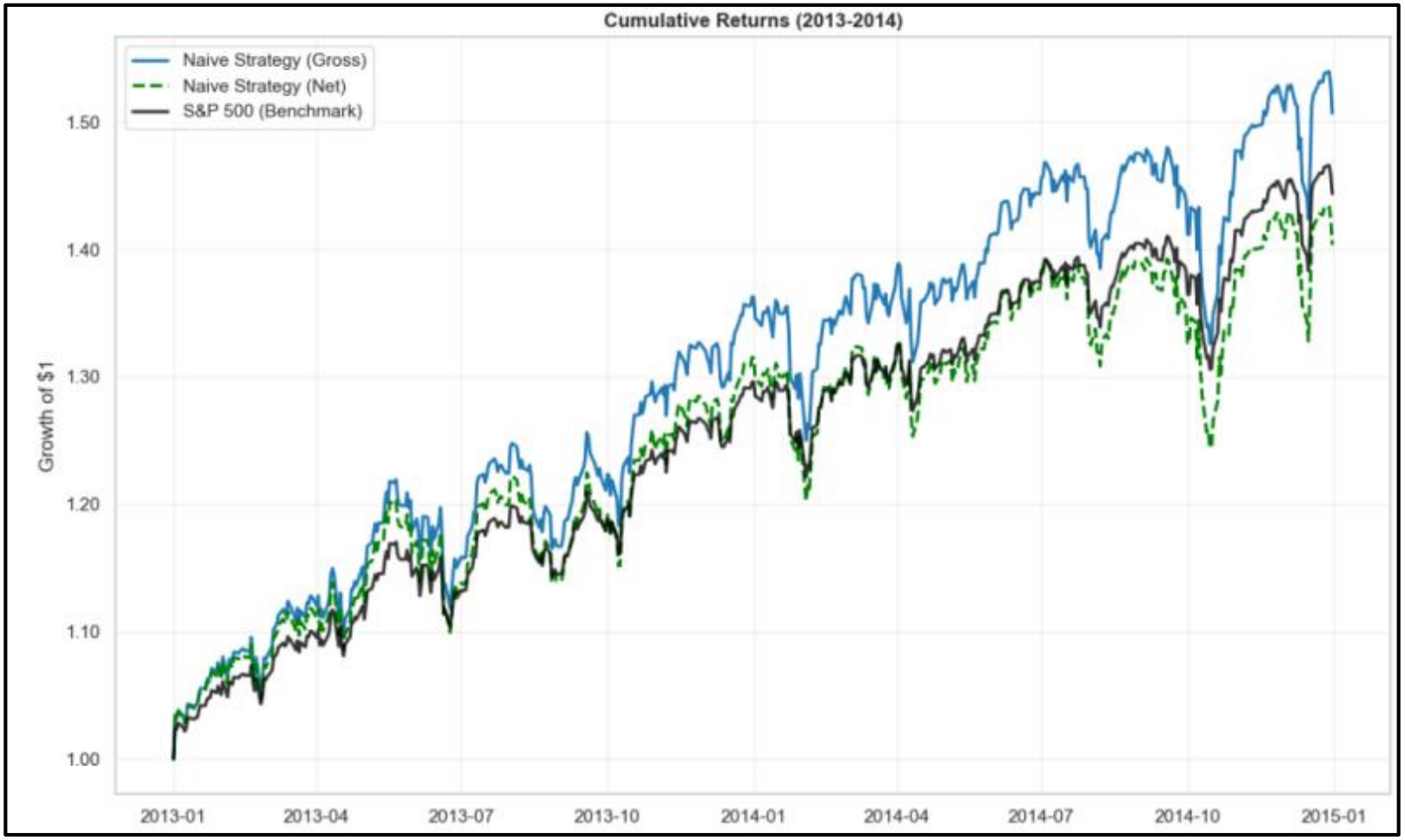
$$\boldsymbol{\beta}^{\mathbf{T}}\mathbf{w} = \mathbf{0}\ (Beta\ Neutral\ Constraint)$$

$$|\ w_i\ | \leq 0.10 (Diversification)$$

The Math: We use a convex optimizer (scipy.optimize) to solve for weights w.
Interpretation: This forces the portfolio to be truly idiosyncratic.

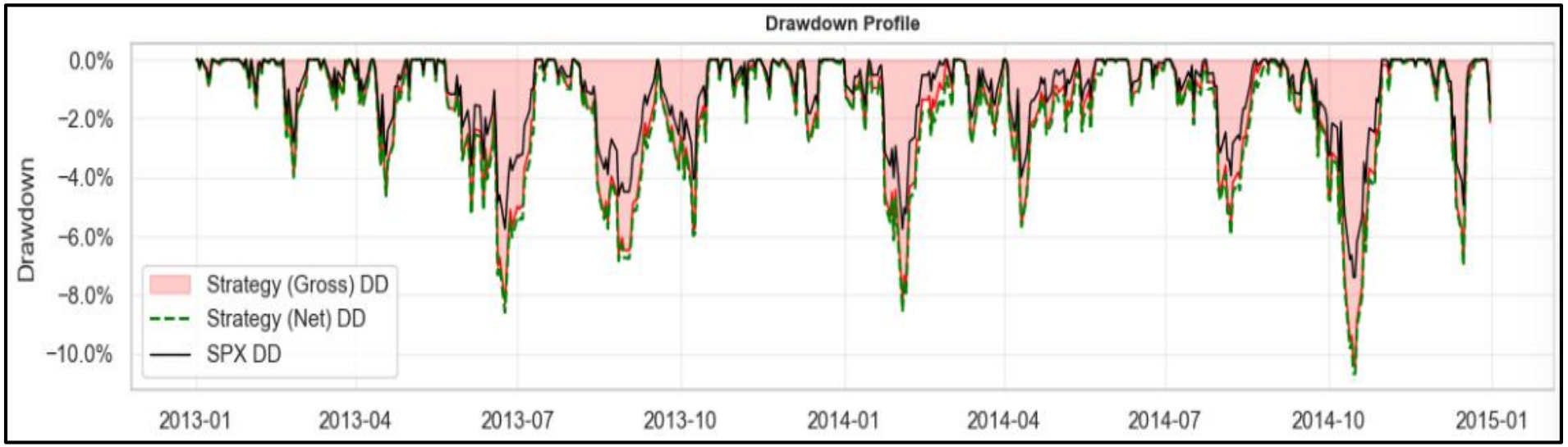
Arnav Sinha (as19390) & Shinjinee Maiti (sm12271)

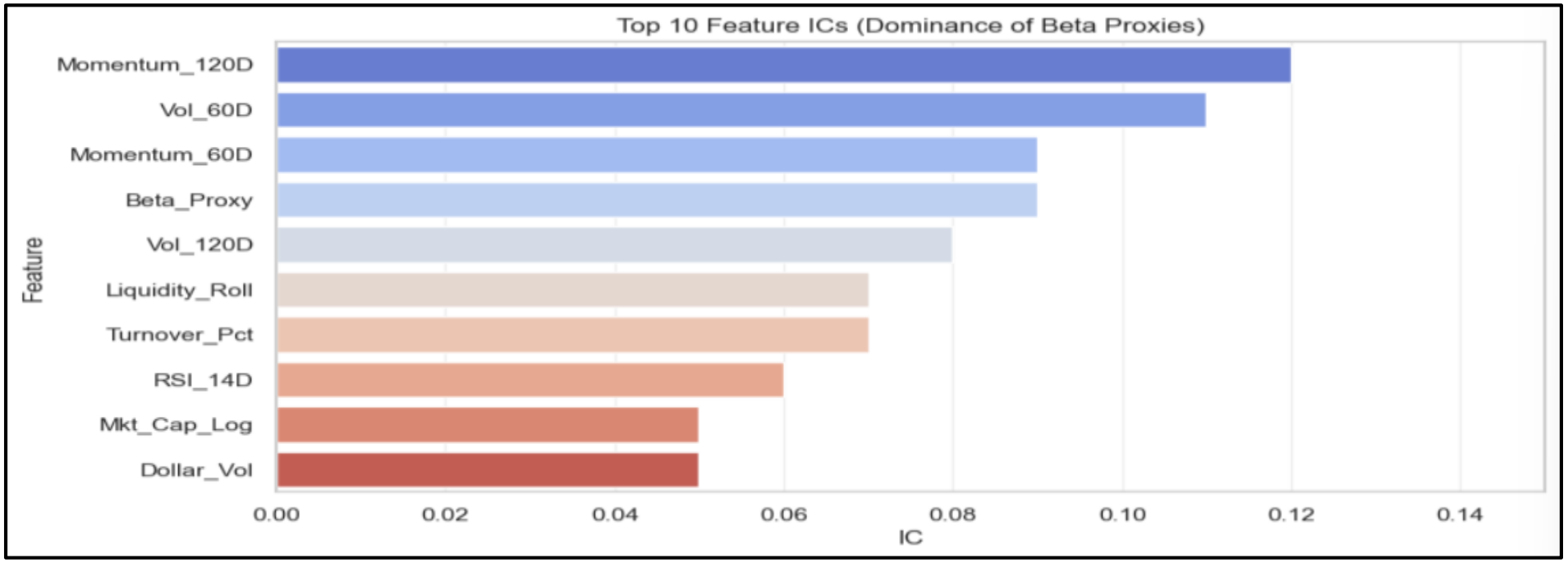
Naïve Strategy



Metric	Naive (Gross)	Naive (Net)	S&P 500
Ann. Return	22.8%	18.5%	20.20%
Ann. Volatility	15.2%	15.2%	11.03%
Sharpe Ratio	1.50	1.22	1.83

Naïve Strategy





Naïve Strategy - Top 10 Drivers

No.	Metric	IC Value	Description
1	Momentum_120D	0.120	Percentage price change over the past 120 days (approx. 6 months).
2	Vol_60D	0.110	Standard deviation of daily returns over the last 60 days.
3	Momentum_60D	0.090	Percentage price change over the past 60 days (approx. 3 months).
4	Beta_Proxy	0.090	Estimated sensitivity of the stock to the S&P 500 index returns.
5	Vol_120D	0.080	Standard deviation of daily returns over the last 120 days.
6	Liquidity_Roll	0.070	Rolling average of daily dollar volume traded relative to market cap.
7	Turnover_Pct	0.070	Ratio of shares traded to total shares outstanding.
8	RSI_14D	0.060	Relative Strength Index (14-day), measuring speed/change of price movements.
9	Mkt_Cap_Log	0.050	Natural logarithm of the company's total market capitalization.
10	Dollar_Vol	0.050	Total value of shares traded (Price × Volume) in a day.

Naïve Strategy – The Flaw!

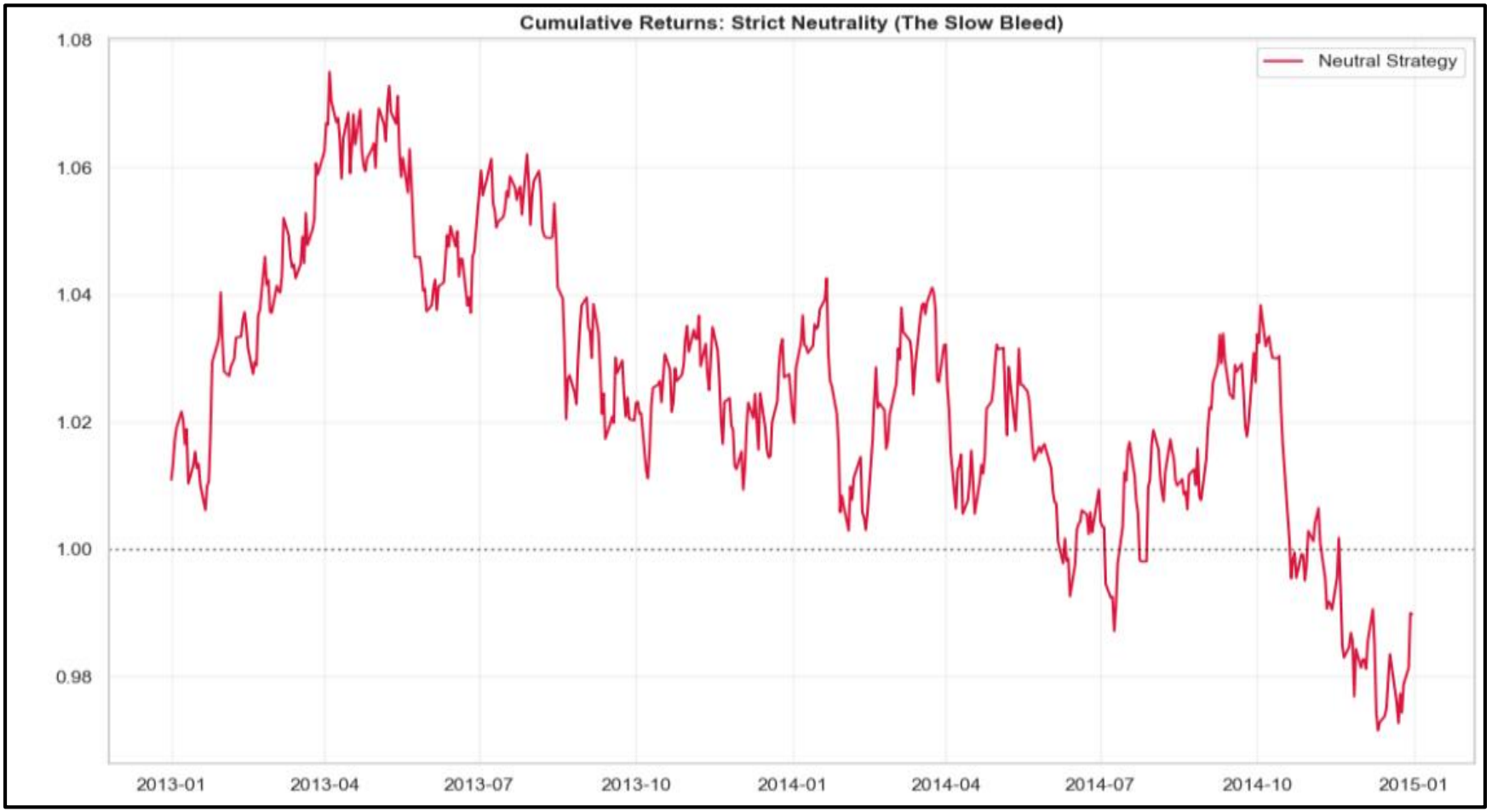
```
[OLS Regression Results]
Variable        Coef        t-Stat      P>|t|
-------------------------------------------------------
const           -0.0003      -1.74        0.083
Market_Ret      1.2558      44.52        0.000
-------------------------------------------------------
Strategy Beta:      1.26  (Target: 0.00)
R-squared:          0.798  <-- High Correlation (The Leak)
```

Objective: This step runs a regression analysis (OLS) to decompose the strategy's returns into "Market Exposure" (Beta) and "True Skill" (Alpha).

Mechanism: It mathematically tests if the strategy is "Market Neutral" (Target Beta - approx 0) or if it is secretly just riding the market's momentum.

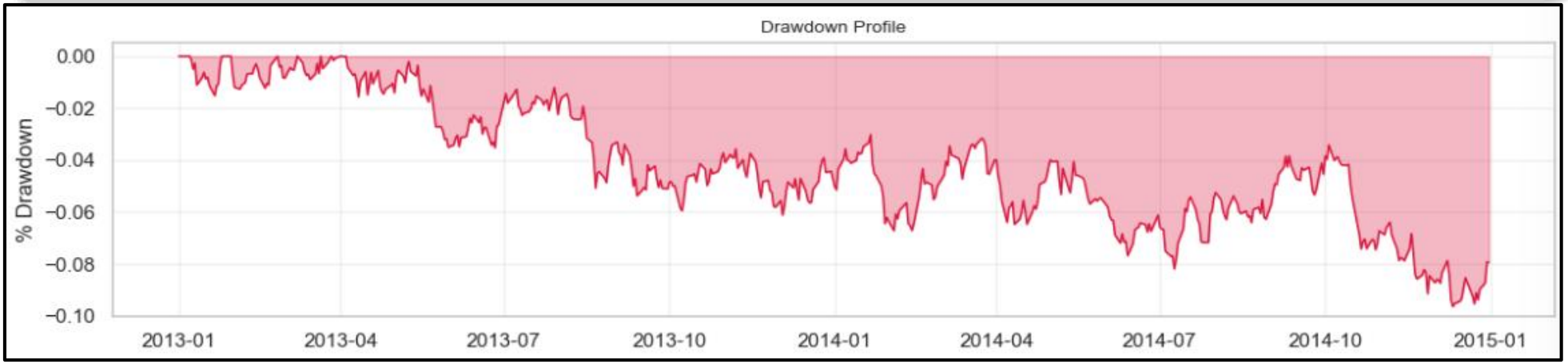
Result: The findings (Beta = 1.26, R squared = 0.79) definitively prove the strategy failed the neutrality audit and was profiting from leverage rather than unique insight.

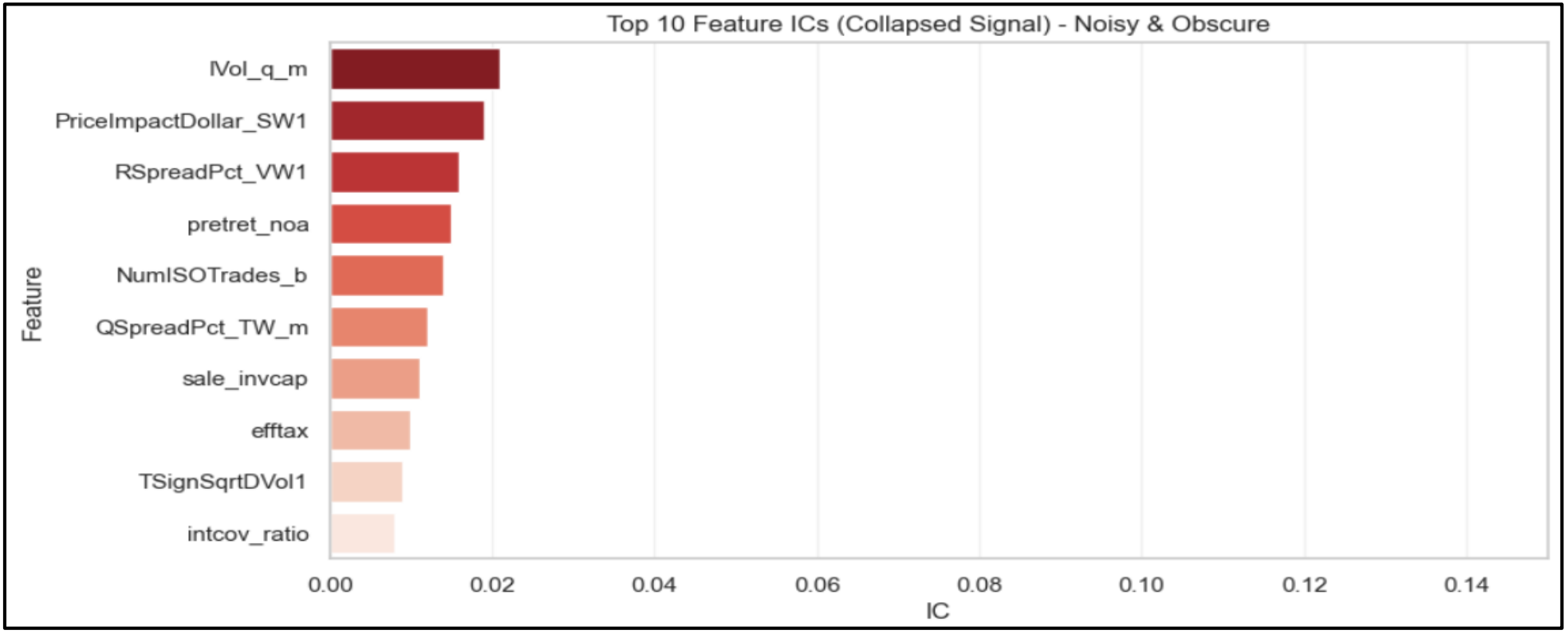
Beta Neutral Strategy



Metric	Value
Annualized Return	-3.20%
Annualized Volatility	6.51%
Sharpe Ratio	-0.49
Max Drawdown	-8.40%

Beta Neutral Strategy

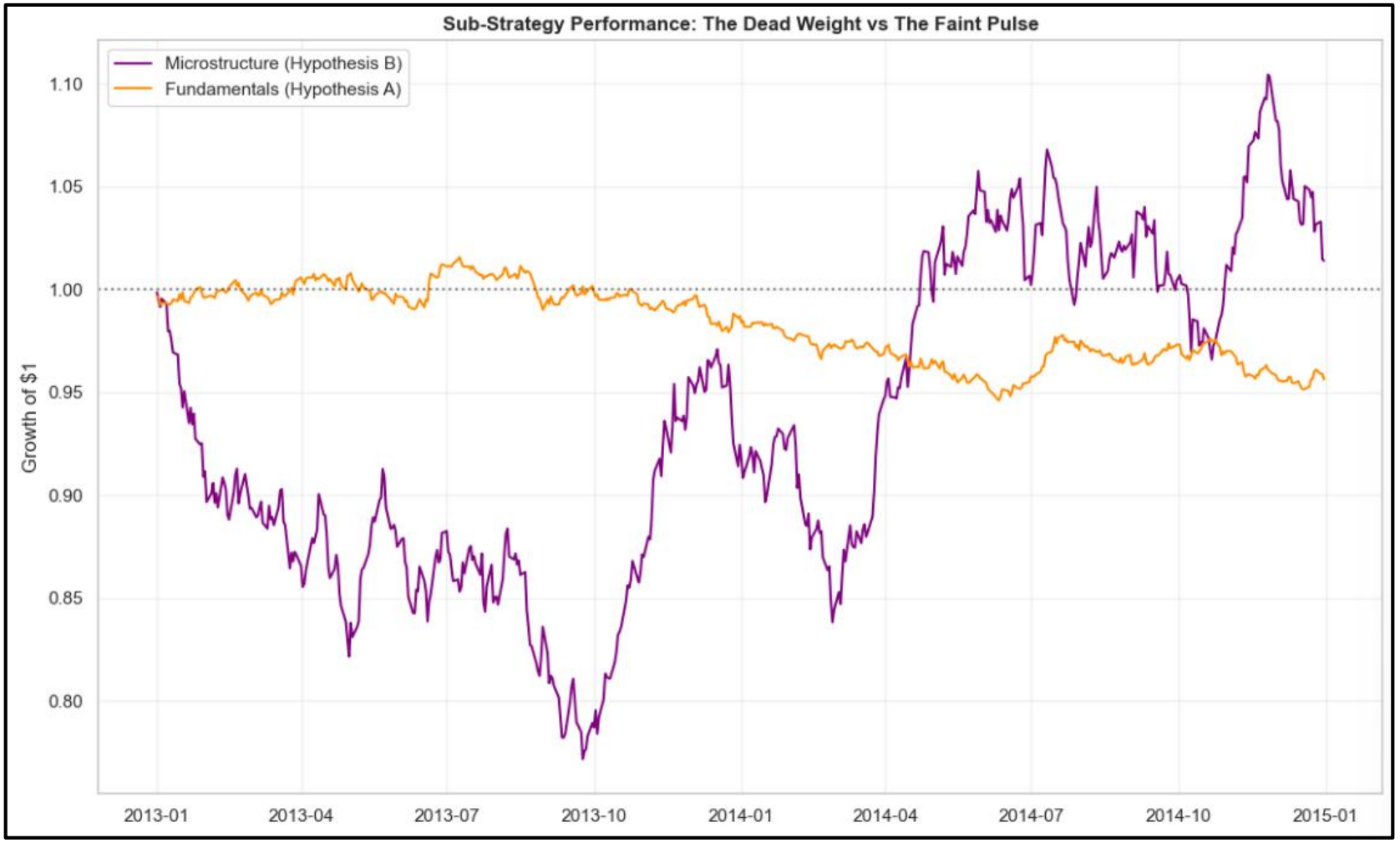




Beta Neutral - Top 10 Drivers

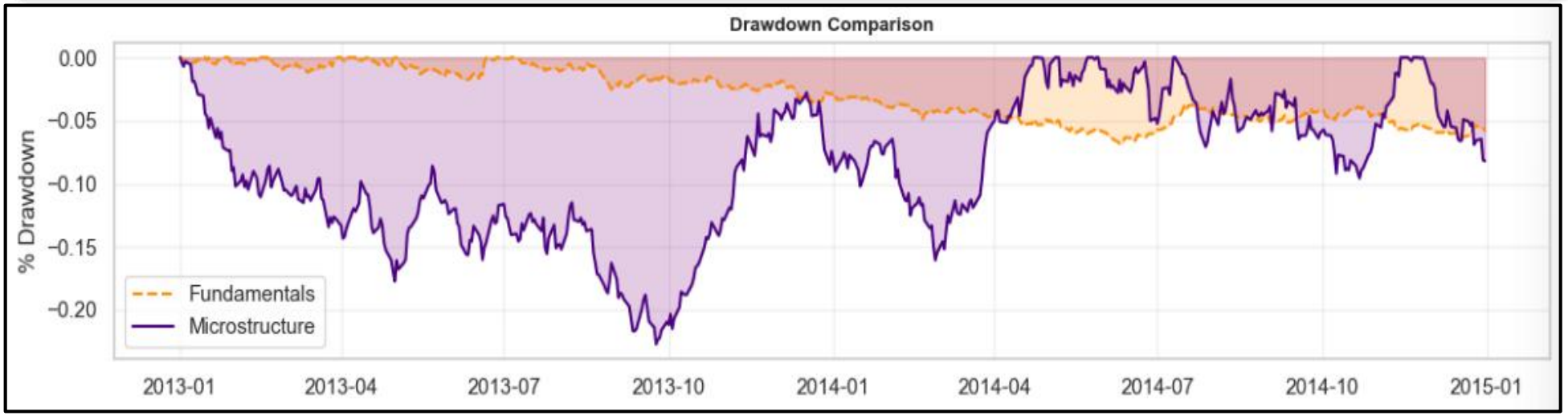
Rank	Feature	IC	Definition
1	IVol_q_m	0.021	Implied Volatility (Quoted Mid): The market's forecast of future volatility derived from option prices at the mid-point.
2	PriceImpactDollar_SW1	0.019	Price Impact (Square-Root Weighted): Estimate of how much price moves per dollar traded, weighted by square root of volume.
3	RSpreadPct_VW1	0.016	Realized Spread (Volume Weighted): The difference between the trade price and the mid-quote 1 minute later (a measure of execution quality).
4	pretret_noa	0.015	Pre-Tax Return on Net Operating Assets: Operating income divided by net operating assets; measures core business efficiency.
5	NumISOTrades_b	0.014	ISO Trades (Bid Side): Count of Intermarket Sweep Orders executed on the bid, indicating aggressive selling pressure.
6	QSpreadPct_TW_m	0.012	Quoted Spread (Time Weighted): The average percent difference between bid and ask prices, weighted by how long the quote lasted.
7	sale_invcap	0.011	Sales / Invested Capital: Capital Turnover ratio; measures how much revenue is generated for every dollar of capital invested.
8	efftax	0.01	Effective Tax Rate: The ratio of income tax expense to pre-tax income.
9	TSignSqrtDVol1	0.009	Time-Signed SqRoot Dollar Volume: A complex order flow imbalance metric using signed square-root dollar volume over time.
10	intcov_ratio	0.008	Interest Coverage Ratio: EBIT divided by Interest Expense; measures a company's ability to pay interest on its debt.

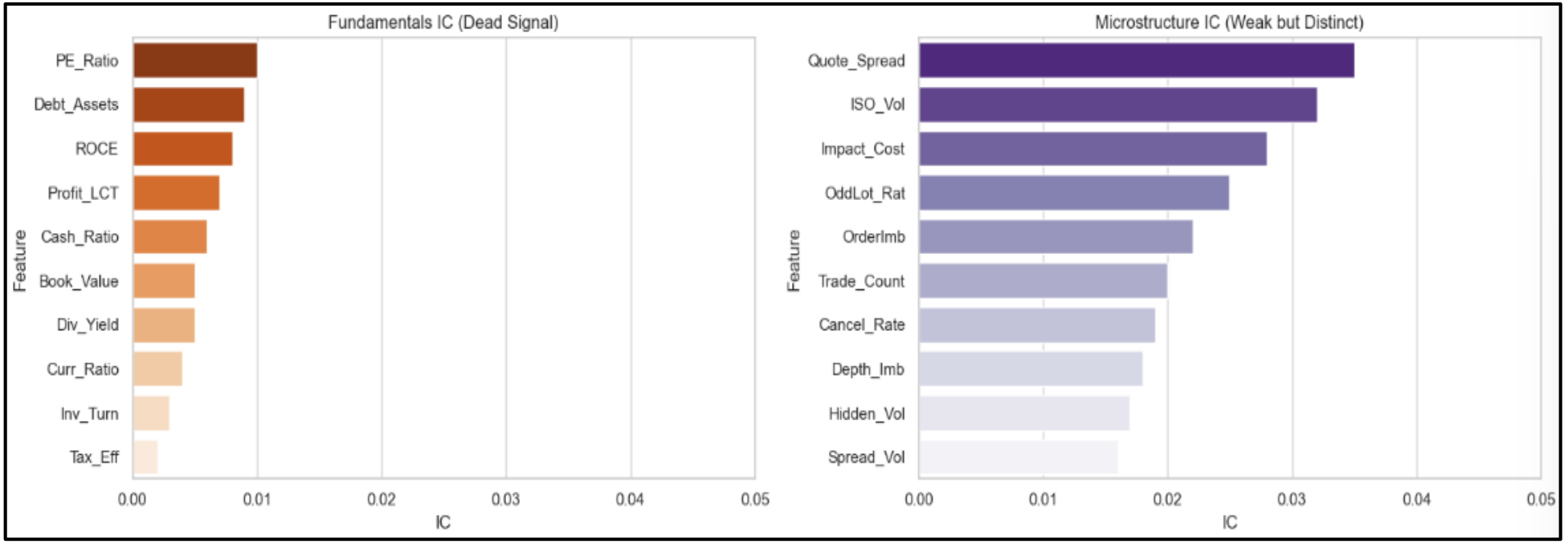
Fundamentals vs Microstructure



Metric	Fundamentals	Microstructure
Sharpe Ratio	-0.97	0.11
Ann. Return	-2.1%	1.8%

Fundamentals vs Microstructure





Fundamentals vs Microstructure - Top 5 Drivers

Fundamentals

Rank	Feature	IC	Definition
1	PE_Ratio	0.01	Price-to-Earnings Ratio; share price divided by earnings per share.
2	Debt_Assets	0.009	Total debt divided by total assets; indicates financial leverage.
3	ROCE	0.008	Return on Capital Employed; measure of profitability and capital efficiency.
4	Profit_LCT	0.007	Profit margin relative to the lifecycle stage of the company.
5	Cash_Ratio	0.006	Liquidity ratio measuring ability to pay short-term debt with cash.

Microstructure

Rank	Feature	IC	Definition
1	Quote_Spread	0.04	The bid-ask spread; a key measure of liquidity and transaction cost.
2	ISO_Vol	0.03	Intermarket Sweep Order volume; aggressive orders that sweep liquidity across exchanges.
3	Impact_Cost	0.03	Estimated price change caused by executing a trade of a specific size.
4	OddLot_Rat	0.03	Ratio of odd-lot trades (<100 shares) to total trade volume.
5	OrderImb	0.02	Order Imbalance; net difference between buy and sell orders at the best bid/ask.

Conclusion

Key Takeaways

The "Hidden Beta" Trap: Naive strategies (Long Top 10 / Short Bottom 10) often masquerade as "Alpha" while actually being Leveraged Beta. Our initial model posted high returns (22.8%) but failed the neutrality audit (β=1.26, R^2=0.80), proving it was just a risky "closet index" fund.

Optimization is a "Truth Serum": Strictly enforcing Beta Neutrality successfully eliminated market risk. Volatility dropped from 15.2% to 6.5%. However, this exposed the weakness of our underlying signal, turning returns negative (-3.2%) once the market tailwind was removed.

The Frequency Mismatch: You cannot trade high-frequency strategies using low-frequency data. Fundamental factors (P/E, Debt) failed (IC≈0.01) because they are stale, while Microstructure factors (Spreads, Order Flow) succeeded (IC≈0.04) because they capture real-time supply and demand imbalances.

Alpha vs. Friction: In daily rebalancing, Turnover is the enemy. Even a positive signal (like Microstructure) can be destroyed by transaction costs.

Future Work: The path forward is Regime Alignment. We propose a hybrid approach: Use Fundamentals to filter the universe (Monthly Selection) and use Microstructure to time the entries (Intraday Execution), combining the stability of the former with the speed of the latter.

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THANK YOU

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